

Your Code

```
1 s=input("enter a string:").lower()
2 freq={}
3 for ch in s:
4     if ch!=" ":
5         if ch in freq:
6             freq[ch] +=1
7     else:
8         freq[ch]=1
9 for ch in sorted(freq):
10    print(ch,"->",freq[ch])
```

Input

Hello World

Output

```
enter a string:  -> 1
```

Your Code

```
1 s=input().lower()
2 v="aeiou"
3 vc=cc=0
4 vf={}
5 cf={}
6 for i in s:
7     if i.isalpha():
8         if i in v:
9             vc+=1
10            vf[i]=vf.get(i,0)+1
11        else:
12            cc+=1
13            cf[i]=cf.get(i,0)+1
14 print(vc,cc)
15 print(max(vf,key=vf.get),vf[max(vf,key=vf.get)])
16 print(min(cf,key=cf.get),cf[min(cf,key=cf.get)])
17 print(round(vc/cc,2))
```

Input

Python Programming

Output

```
4 13
o 2
y 1
0.31
```

Your Code

```
1 def rle_compress(s: str) -> str:
2     if not s:
3         return ""
4
5     compressed = []
6     i = 0
7     n = len(s)
8
9     while i < n:
10        ch = s[i]
11        count = 1
12        while i + 1 < n and s[i + 1] == ch:
13            count += 1
14            i += 1
15
16        if count == 1:
17            compressed.append(ch)
18        else:
19            compressed.append(ch + str(count))
20        i += 1
21
22    compressed_str = "".join(compressed)
23    return compressed_str if len(compressed_str) < len(s) else s
```

Input

aabbccdddee

Output

Compressed: a2b3c2d4e2
Decompressed: aabbccdddee

Your Code

```
1 import re
2 from collections import Counter
3
4 STOP_WORDS = {
5     "the", "is", "a", "an", "in", "of", "and", "to", "for"
6 }
7
8 def extract_sentences(text: str):
9     # Sentence ends with . ! or ?
10    # Capture sequences ending with those punctuation marks.
11    sentences = re.findall(r'^[!?!?]+[.!?]', text)
12    # Handle case where text has no ending punctuation:
13    remainder = re.sub(r'^[!?!?]+[.!?]', '', text)
14    if remainder.strip():
15        # If there's leftover without proper ending, treat it as a
16        sentences.append(remainder.strip())
17    return [s.strip() for s in sentences if s.strip()]
18
19 def tokenize_words(text: str):
20    # Remove punctuation attached to words; keep alphabetic words a
21    # e.g., don't -> dont (or keep as dont). We'll treat it as a wo
22    words = re.findall(r'[A-Za-z]+(?:'[A-Za-z]+)?', text.lower())
23    # Normalize "don't" -> "dont"
```

Input

hi

Output

Words: 1 | Sentences: 1 |
Paragraphs: 1
Avg Word Length: 2.0 | Avg
Sentence Length: 1.0
Top 5 Words: hi(1)
Flesch Reading Ease: 121.2